ArcGIS Earth Object Factory 2.0 - User Manual

Installing Object Factory - read this first

1. **Extract All first:** right-click the downloaded zip > **Extract All**, then open the new folder. (Double-clicking files while they are still inside the zip does not work.)
2. **Double-click** `Run_Object_Factory.bat` - the only .bat in the folder. This manual sits right next to it. The **System Files** folder holds the program - leave it alone.
3. **Blue "Windows protected your PC" box?** Click **More info**, then **Run anyway**. It only asks the first time. (Or, before extracting: right-click the zip > **Properties** > tick **Unblock** > **OK**.)
4. **Python - nothing to install by hand.** If Python is missing, the window asks **Install Python now? [Y/N]**: type **Y** and press Enter. It installs just for you (no admin needed). On an older PC this can take several minutes and Windows Defender may make the PC look frozen - **do not close the window**.
5. To see your objects you need **ArcGIS Earth** (free from Esri: https://www.esri.com/en-us/arcgis/products/arcgis-earth/downloads): drag the KMZ it makes onto the map.

Something stops it? The window says what to do in plain words, and keeps a log in `System Files\Ready_To_Fly_log.txt`. On a work computer that blocks installs, ask IT to install Python 3 from python.org.

2.0 (2026-11-08): the whole ArcGIS Earth Unlocked family moved to 2.0 as one reference point. Objects come out exactly as in v1.3; the program file is now `Object_Factory_v2.0.py`.

v1.3 (2026-11-06): a tidy folder - `Run_Object_Factory.bat` plus `System Files`. Nothing else changed.

v1.2 (2026-11-02): new **Hide above (ft)** setting (section 4) so objects stop cluttering the view from high above; the Objects tab takes **Altitude in feet**; an Info popup with nothing in it now shows the object's name. See OBJECT_FACTORY_V1.2_CHANGES.md. v1.1 (2026-10-31) added the coordinate readback, overwrite prompt and safe writes (OBJECT_FACTORY_V1.1_CHANGES.md).

A printable copy of this manual is available as ArcGIS_Earth_Object_Factory_v2.0_User_Manual.pdf.

Create geographic objects without hand-writing KML.

Coordinates + PNG → Object Factory → KMZ/KML → ArcGIS Earth

v1.0 was commissioned on October 29, 2026. Airborne PNG objects, batch object sets, the N/E/S/W Sky Compass and the Beacon were live-proven in ArcGIS Earth. v1.1 (October 31, 2026) was a hardening release. v1.2 (November 2, 2026) adds Hide above and altitude in feet.

> **Core PNG rule:** the original PNG is never modified. Packaged icon copies are normalized to a maximum dimension of 256 px while preserving aspect ratio and transparency.

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1. Quick Start

Object Factory is a Windows Python/Tkinter application. Normal use does not require ArcGIS Pro, QGIS, Pillow, or other third-party Python packages.

What you need

- Windows with Python 3, including Tkinter - since 1.3.1 `Run_Object_Factory.bat` installs Python for you if it is missing (answer **Y**; no admin needed).
- ArcGIS Earth desktop to view the generated KML/KMZ files.
- PNG images only when you want custom icons. Preset icons are included.

Start the factory

1. Download `ArcGIS_Earth_Object_Factory_v2.0.zip` and extract it to a normal folder. It contains an `Object_Factory_v1.3.1` folder.
2. Inside are just three things: `Run_Object_Factory.bat`, this manual as a PDF, and `System Files`. Keep them together.
3. Double-click `Run_Object_Factory.bat`.
4. Choose **Objects**, **Batch**, **Sky Compass**, or **Beacon**.
5. Create the output and open the resulting KML/KMZ in ArcGIS Earth.

What's in the zip



Fastest first test

Open **Objects**, choose the **firetruck** preset, enter a known coordinate, leave labels off, add short Info text, press **CREATE OBJECT KMZ**, and open the KMZ (saved to your Desktop by default) in ArcGIS Earth. This mirrors the live-proven first Object Factory primitive.

Main package files

- `Run_Object_Factory.bat` - Windows launcher (the one to double-click).
- `System Files\` - everything below:
- `Object_Factory_v2.0.py` - main application.
- `icons/` - 11 object presets plus N/E/S/W compass PNGs.
- `sample_objects.txt` - full-format batch example.
- `simple_objects.txt` - shorthand batch example.
- `SAMPLE_Object_Set.kmz`, `SAMPLE_Linked_Camera.kmz`, `SAMPLE_Sky_Compass.kmz`, `SAMPLE_Beacon.kml` - ready-made outputs to open in ArcGIS Earth (also downloadable one by one from the GitHub page).

Every output path has a **Browse** button. By default, files are saved to your Desktop.

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2. Main Window and Coordinate Entry

The factory uses one object-generation engine behind four task tabs: **Objects**, **Batch**, **Sky Compass** and **Beacon**. A status line at the bottom reports what the factory just did.

Coordinate entry

Manual - type decimal latitude and longitude directly.

From Clipboard - reads coordinates copied from Google Maps or other text. Google Maps' `30.40098, -81.93471` (latitude first) is the reference format. N/S/E/W letters (`30.4009768°N 81.9347098°W`) and degrees-minutes-seconds (`30°24'03.5"N 81°56'05.0"W`) are also understood. The line under the button states exactly what was read, e.g. `Read as Latitude 30.400980 N, Longitude 81.934710 W`.

Google Maps workflow

1. Right-click a location in Google Maps.
2. Click the coordinate pair to copy it.
3. Return to Object Factory and press **From Clipboard**.
4. Check the `Read as Latitude ... Longitude ...` line.
5. Create the object.

Latitude must be between -90 and 90. Longitude must be between -180 and 180. The factory validates these ranges before output is built.

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3. Objects Tab - Create One Geographic Object

Use **Objects** for one floating or ground-level PNG object. The factory creates one KMZ containing the KML and the normalized icon copy.

Typical workflow

1. Choose a preset or **Browse** to your own PNG. The PNG inspection box shows its size and what will be packaged.
2. Enter the object name and coordinate.
3. Set altitude (feet), altitude mode, and icon scale.
4. Optionally set **Hide above (ft)** so the object disappears when you are far away (section 4).
5. Normally leave **Show label** off.
6. Turn on **Show tether** only when you want a visible line to the ground.
7. Choose **Info** or **None** for popup behavior.
8. Enter optional Info text and hyperlink.
9. Choose the output KMZ path (default `ArcGIS_Earth_Object.kmz` on your Desktop). If the file already exists, the factory asks before replacing it.
10. Press **CREATE OBJECT KMZ**.

Field reference

- **Preset** - loads a built-in icon and starting altitude/scale. All fields remain editable.
- **Object name** - KML Placemark name.
- **Latitude / Longitude** - geographic anchor point.
- **PNG icon** - preset or custom transparent PNG. Source is never modified.
- **Altitude (ft)** - altitude in feet (converted to metres inside the KMZ). Presets fill in their altitude rounded to 10 ft.
- **Altitude mode** - `relativeToGround`, `absolute`, or `clampToGround`.
- **Icon scale** - KML icon scale.
- **Popup** - `Info` or `None`. With `Info` and no Info text or link, the popup shows the object's name.
- **Hide above (ft)** - the object shows only while you are within about this distance; blank = always visible. See section 4.
- **Show tether** - adds a vertical line from ground to the object.
- **Show label** - continuously displays the object name. Default is off.
- **Info text** - short text shown after clicking.
- **Link text / URL** - optional standard hyperlink in the KML information.
- **Output KMZ** - output path.

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4. PNG Handling, Altitude, Hide Above, Labels, and Links

Automatic PNG normalization

When you select a PNG, the factory inspects its dimensions and packaged size. Oversized source images are reduced for the KMZ. Smaller clean icons are not needlessly enlarged.

Locked behavior:

- Maximum packaged dimension: **256 px**.
- Aspect ratio is preserved.
- Transparency is preserved.
- The source PNG remains byte-for-byte untouched.
- A wide helicopter stays wide; a tall object stays tall.
- A clean 128 × 128 icon may remain 128 × 128.
- Batch mode reuses a normalized icon rather than packing duplicate copies unnecessarily.

Altitude modes

- **relativeToGround** - altitude above terrain. Normal choice for floating icons and sky objects.
- **absolute** - absolute elevation. Use only when intentional.
- **clampToGround** - places the object on the ground surface.

Hide above (ft)

Many objects over a city turn into a pile of giant icons when you are high above it. **Hide above** fixes that: type a distance in feet, and the object shows only while you are within about that distance of it.

- **Looking straight down,** the number is the **eye alt** ArcGIS Earth shows in the bar at the bottom of the map. `3000` = the object shows below about 3,000 ft eye alt.
- **Flying low,** you still see the object off in the distance, out to roughly the same distance (measured: about a third farther sideways).
- **Blank** = always visible.

Suggested starting values: a few thousand feet for street-level objects (hydrants, cameras at an intersection); 20,000-40,000 ft for objects you want to see across a county. Adjust after a look in ArcGIS Earth.

In ArcGIS Earth the object appears inside a folder of its own name in My Data; that folder carries the Hide above setting. Popups and links work as normal. The cutoff is approximate and was measured on a full-screen ArcGIS Earth window; a smaller window hides objects a little sooner.

Labels and click information

The project default is **no permanent label** for most objects. The icon carries the first layer of meaning; click only when more information is needed.

> **See it → recognize it → click it → understand it.**

Hyperlinks

Enter link text such as `OPEN LIVE CAMERA`, `STREET VIEW`, `FACILITY PAGE`, or `OPEN MAP`, then enter a normal URL. The factory writes a standard HTML hyperlink into the KML description.

The factory writes the link structure; browser-launch behavior depends on ArcGIS Earth and Windows/browser configuration.

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5. Preset Library

Presets are convenience starting points. Every preset feeds the same generic PNG-object engine.

Preset	Default altitude (Objects tab / Batch)	Scale	Typical meaning
object	490 ft / 150 m	1.4	generic object / pin
helicopter	980 ft / 300 m	1.5	aviation marker
fuel	490 ft / 150 m	1.4	fuel stop
bird	390 ft / 120 m	1.4	lesson / wildlife object
hazard	660 ft / 200 m	1.5	warning / hazard
home	660 ft / 200 m	1.5	home / base
staging	490 ft / 150 m	1.5	staging point
water	490 ft / 150 m	1.5	water source
firetruck	490 ft / 150 m	1.5	apparatus / fire-service point
camera	590 ft / 180 m	1.5	viewpoint / camera
medical	590 ft / 180 m	1.5	hospital / medical point

In batch lists, these alternative names also work: `pin` (object), `heli` (helicopter), `gas` / `gas_pump` (fuel), `warning` (hazard), `base` (home), `truck` (firetruck), `viewpoint` (camera), `hospital` (medical). An unrecognized type becomes a generic `object`.

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6. Batch Tab

Use **Batch** to create many objects in one KMZ. This is the fastest path for classrooms, training scenarios, operational object packs, or any repeated coordinate list.

Batch workflow

1. **Browse** to the object list (a `.txt` file). The default is the included `sample_objects.txt` (in `System Files`).

2. Press **VALIDATE LIST**. The summary line reports how many objects were read, or names the line with a problem.
3. Choose the output KMZ (default `Object_Set.kmz` on your Desktop).
4. Press **CREATE OBJECT SET**.

Two list formats are supported:

Shorthand format

```
type,lat,lon[,alt_m][,name]
```

```
fuel,30.410104,-81.922396
heli,30.398210,-81.931440,300
bird,30.402778,-81.918620,120,Mystery Bird
home,30.385525,-81.930807
```

The preset supplies icon, normal altitude, and scale when omitted. Without a name, the object is named after its type.

Full pipe-delimited format

```
type|name|lat|lon|alt_m|icon|description|url|link_text|scale|altitude_mode|tether|pop
up|label|hide_above_ft
fuel|Fuel Stop|30.410104|-81.922396|150||Use this fuel
stop.|||1.4|relativeToGround|no|Info|no|
camera|Viewpoint|30.394120|-81.927300|180||Another way to see this
place.|https://www.google.com/maps|OPEN MAP|1.5|relativeToGround|no|Info|no|5000
```

Blank lines and lines beginning with `#` may be used for readability and are ignored. The `hide_above_ft` column is optional; lists without it work as before.

The integrated self-test generated a **100-object batch KMZ** and verified duplicate-icon reuse.

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7. Full Batch Field Reference

- `type` - preset name or alias.
- `name` - object name.
- `lat` / `lon` - coordinate.
- `alt_m` - altitude in **metres** (the Objects tab uses feet; batch lists keep metres so existing lists still work).
- `icon` - optional custom PNG path. Blank uses the preset icon; a relative path is looked up beside the list file.

- `description` - Info text.
- `url` - optional URL.
- `link_text` - hyperlink text.
- `scale` - KML icon scale.
- `altitude_mode` - `relativeToGround`, `absolute`, or `clampToGround`.
- `tether` - yes/no, true/false, 1/0, on/off.
- `popup` - `Info` or `None`.
- `label` - yes/no. Normally no.
- `hide_above_ft` - optional Hide above distance in feet (section 4). Blank = always visible.

Use `sample_objects.txt` for the full format and `simple_objects.txt` for the shorthand format.

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8. Sky Compass

Sky Compass creates four giant geographic PNG markers - **N, E, S, W** - around a center point. It was designed for EarthStick horizon flight, where the letters act as persistent orientation landmarks in the sky.

Default starting values:

- Radius: **100 km**.
- Altitude AGL: **20,000 m**.
- Icon scale: **6**.
- Output: one `Sky_Compass.kmz` containing all four markers.

Create a Sky Compass

1. Set the center manually or with **From Clipboard**.
2. Set the radius.
3. Set altitude AGL.
4. Set icon scale.
5. Choose the output KMZ (default `Sky_Compass.kmz` on your Desktop).
6. Press **CREATE SKY COMPASS** and open the KMZ in ArcGIS Earth.

The POF3 N/E/S/W geometry was live-tested successfully. Radius, altitude, and scale remain adjustable because the best values depend on the operating area and preferred view.

> **Sky Compass = orientation. Earth Beacon = destination. EarthStick = movement.**

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9. Beacon Tab

Beacon preserves the accepted Earth Beacon geometry: a tall vertical KML line anchored to a geographic point so the destination remains visible from long range in an oblique/horizon view.

Standard starting values

- Height: **100 km**.
- Line width: **6**.
- Default color: **Red**.
- Available colors: Red, Yellow, Cyan, Green, Magenta, White, Orange.
- Beacon name: **EarthStick Beacon** (shown as the label at the base).
- Output: KML (default `EarthStick_Beacon.kml` on your Desktop).

Standard workflow

1. Enter a beacon name and the destination coordinate, manually or with **From Clipboard**.
2. Keep 100 km height and width 6 for the accepted baseline unless you intentionally want another presentation.
3. Choose color.
4. Press **CREATE BEACON KML**.
5. Open the KML in ArcGIS Earth.

Earth Beacon geometry is already live-tested and accepted. The Factory tab preserves it rather than redesigning it.

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10. Using Factory Output in ArcGIS Earth

- KMZ/KML objects appear under **My Data**.
- Toggle the whole layer on/off from the Data panel.
- Most factory objects are intended to show an icon without a permanent label.
- Click an **Info** object to open its information.
- Objects with links expose the hyperlink through KML information content.
- ArcGIS Earth can edit KML object information after creation through object Properties / Info.

A generated object is therefore not locked. You can edit its Rich Text or HTML information in ArcGIS Earth without rebuilding the foundation map.

> **Offline object = useful by itself. Connected object = potentially unlimited extension.**

Camera / Viewpoint objects

A camera object is a geographic camera PNG plus an optional link. It can point to a live webcam, Street View, photo page, video, or normal web resource. A separate always-on-top mini Camera Viewer is a future concept and is **not** part of the factory.

Native popup size

ArcGIS Earth presents KML information in a relatively large native panel. This is viewer behavior. Keep Info text concise or choose `Popup = None` for purely visual objects.

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11. Troubleshooting and Verification

Factory does not start

Confirm Python 3 is installed and includes Tkinter. Run the BAT from the extracted folder so the `icons` folder remains beside the Python script.

PNG not ready

Confirm the file is a readable PNG. The factory intentionally does not require Pillow and does not accept other icon formats.

Object too small or large

Adjust **Icon scale** and recreate the KMZ.

Object disappears too early or too late

Raise **Hide above (ft)** if it disappears too soon; lower it if it clutters the view too long. Leave it blank for always visible.

Object too high or low

Check **Altitude** and **Altitude mode**. `relativeToGround` is the normal floating-object setting.

Large popup

Use concise Info text or `Popup = None`. Popup size is an ArcGIS Earth presentation behavior.

Batch validation problem

The message names the line (and, for header lists, the object), e.g. `Line 4: latitude '30.40' is not a number.` Line numbers count every line in the file, including comments and blank lines. Also check the delimiter, header spelling (`lat` and `lon` columns), icon paths, and that shorthand lines contain at least type, latitude, and longitude.

Clipboard coordinates not accepted

The message explains why. Copy just one coordinate pair; extra numbers such as `Pin 1:` are rejected. If necessary, enter the coordinate manually.

Operational verification checklist

1. Create a known firetruck and confirm placement, Info text, and click behavior.
2. Select an oversized PNG and verify the source remains unchanged.
3. Create an object with a harmless URL and confirm the link appears.
4. Choose `Popup = None` and inspect the result.
5. Load `sample_objects.txt` and inspect mixed objects.
6. Load `simple_objects.txt` and inspect the shorthand objects. 6a. Create an object with Hide above 3000, climb straight over it and check it disappears near 3,000 ft eye alt.
7. Recreate the Sky Compass and inspect N/E/S/W behavior.
8. Create a standard 100 km red Beacon and confirm accepted geometry remains unchanged.

Developer self-test

From a command prompt in the `System Files` folder:

```
py -3 Object_Factory_v2.0.py --selftest
```

The self-test covers PNG normalization/original preservation, single-object KMZ creation, hyperlink output, no-popup KML style, a 100-object batch, duplicate-icon reuse, shorthand parsing, Sky Compass generation, Beacon generation, and (v1.2) Hide above and name popups.

> **Release rule:** Object Factory v1.2 is the stable accepted baseline. Future functional changes receive a new version and should preserve known-good behavior unless the change specifically targets it.

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12. Operating Philosophy

Object Factory is deliberately not a GIS editor. It is a simple object-authoring bridge into ArcGIS Earth.

> **Find a place → copy coordinates → choose an object → create it → open ArcGIS Earth.**

Project rules:

- Foundation maps stay stable; small information layers change.
- The icon carries the first layer of meaning.
- Most objects stay unlabeled until clicked.
- Original PNGs are never modified.
- Offline capability comes first; online links add an optional second layer.
- Presets simplify common tasks, but all presets use the same generic object engine.
- The sky is a programmable information layer, not empty background.

> **See it → recognize it → click it → understand it.**
>
> **The world is already built. We just add the objects.**

Related Project Family

Object Factory is part of **ArcGIS Earth Unlocked**:

https://github.com/Jim-dc95811/ArcGIS-Earth-Unlocked

The umbrella connects the offline-map foundation, EarthStick navigation, Object Factory authoring, PointLayer public point data, and Parcel100 parcel intelligence while keeping each specialized tool independent.